

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Branch: CSM

Section: B

Task No.: 3

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1. TITLE

Design a Responsive Webpage Using HTML5 and CSS3

2. AIM

To design and develop a modern, responsive webpage using HTML5 and CSS3 with a navigation bar, hero section, about section, services section, testimonials section, contact form, and footer.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 webpage.
- To design a webpage using HTML5 semantic elements.
- To apply CSS3 styling to HTML elements.
- To create a responsive navigation bar with a sticky header.
- To design an attractive hero section using CSS gradients.
- To create service and testimonial cards using CSS Flexbox.
- To design and style a contact form.
- To implement hover effects, smooth scrolling, and scroll-based reveal animations.
- To make the webpage responsive using CSS media queries.
- To understand the separation of HTML structure and CSS presentation.

4. THEORY

HTML5 provides the structural elements of a webpage — headings, sections, forms, and text content. CSS3 is used to style and lay out that structure — colours, spacing, fonts, and responsive behaviour — while keeping the HTML file focused purely on content and the CSS file focused purely on presentation. Keeping them separate makes a site easier to maintain and reuse.

Several CSS3 features, along with a small amount of JavaScript, make this page attractive, interactive, and responsive.

- Flexbox (display: flex) arranges the navigation links, service cards, and testimonial cards in a row that wraps neatly onto new lines when space runs out.
- Media queries (@media) apply different CSS rules at different screen widths, which is the core mechanism behind responsive design.
- Transitions and :hover create smooth colour, shadow, and lift changes when the user points at a link, button, or card.

- scroll-behavior: smooth makes in-page anchor links (like the nav bar) scroll smoothly instead of jumping instantly.
- position: sticky keeps the header fixed at the top of the viewport, and a short JavaScript scroll listener adds a shadow once the page is scrolled.
- The IntersectionObserver API (JavaScript) detects when a section or card enters the viewport and fades/slides it into view.
- Box model properties (padding, margin, border-radius, box-shadow) control spacing and give the cards and buttons their rounded, elevated look.

The basic flow of the page is:

Browser loads index.html → Browser links week-3.css → CSS rules are applied to HTML elements → JavaScript attaches scroll and intersection listeners → Layout adjusts to screen width via media queries

5. ALGORITHM / PROCEDURE

1. Create two files: index.html and week-3.css, and link the CSS file inside the HTML head.
2. Reset default margins/padding and set box-sizing to border-box for consistent sizing.
3. Build a sticky header with a logo and a navigation list linking to in-page sections.
4. Create a hero section with a gradient background, heading, description, and a call-to-action button.
5. Create an about section with a heading and a descriptive paragraph.
6. Create a services section and arrange three service cards using CSS Flexbox.
7. Create a testimonials section and arrange three testimonial cards using CSS Flexbox.
8. Style each card with padding, rounded corners, and a shadow that lifts on hover.
9. Create a contact form with labelled text, email, and message fields.
10. Style the form inputs with borders, rounded corners, and a focus colour change.
11. Create a footer with a centred copyright notice.
12. Add scroll-behavior: smooth and a JavaScript scroll listener so the header gains a shadow when scrolled.
13. Add a JavaScript IntersectionObserver so sections and cards fade into view as the user scrolls.
14. Add media queries at 768px and 480px breakpoints to adjust the layout for tablets and phones, then open index.html in a browser and resize the window to verify responsive behaviour.

6. PROGRAM

```

• • •   index.html
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>PixelForge Studio</title>
7    <link rel="stylesheet" href="week-3.css">
8  </head>
9  <body>
10
11  <!-- ===== HEADER / NAV ===== -->
12  <header id="site-header">
13    <nav>
14      <div class="logo">PixelForge</div>
15      <ul class="nav-links">
16        <li><a href="#home">Home</a></li>
17        <li><a href="#about">About</a></li>
18        <li><a href="#services">Services</a></li>

```

```

19         <li><a href="#testimonials">Testimonials</a></li>
20         <li><a href="#contact">Contact</a></li>
21     </ul>
22 </nav>
23 </header>
24
25 <!-- ===== HERO SECTION ===== -->
26 <section class="hero" id="home">
27     <h1>Ideas Deserve Great Design</h1>
28     <p>PixelForge Studio turns rough concepts into polished digital products – websites,
29     apps, and brands that people remember.</p>
30     <a href="#contact" class="btn">Start a Project</a>
31 </section>
32
33 <!-- ===== ABOUT SECTION ===== -->
34 <section class="about" id="about">
35     <h2>Who We Are</h2>
36     <p>PixelForge Studio is a small, focused creative team that partners with startups and
37     growing businesses to design and build products worth using. We obsess over the details –
38     layout, motion, speed, and clarity – so your users never have to think twice.</p>
39 </section>
40
41 <!-- ===== SERVICES SECTION ===== -->
42 <section class="services" id="services">
43     <h2>What We Do</h2>
44     <div class="service-cards">
45         <div class="card">
46             <div class="card-icon">🎨</div>
47             <h3>Brand & UI Design</h3>
48             <p>Distinct visual identities and interfaces that reflect who you are, not a
49             generic template.</p>
50         </div>
51         <div class="card">
52             <div class="card-icon">💻</div>
53             <h3>Web & App Development</h3>
54             <p>Fast, reliable builds using clean code – from landing pages to full product
55             dashboards.</p>
56         </div>
57         <div class="card">
58             <div class="card-icon">📈</div>
59             <h3>Growth & Optimization</h3>
60             <p>SEO-friendly structure, performance tuning, and conversion-focused layouts
61             that actually convert.</p>
62         </div>
63     </div>
64 </section>
65
66 <!-- ===== TESTIMONIALS SECTION ===== -->
67 <section class="testimonials" id="testimonials">
68     <h2>Client Stories</h2>
69     <div class="testimonial-cards">
70         <div class="testimonial-card">
71             <p class="quote">"PixelForge rebuilt our entire product page and bookings jumped
72             almost immediately. They actually listen."</p>
73             <p class="author">– Priya Nair, Founder, Trailmate</p>
74         </div>

```

```

68     <div class="testimonial-card">
69         <p class="quote">"Clear communication, sharp design instincts, and they delivered
        ahead of schedule. Rare combination."</p>
70         <p class="author">– Karthik Iyer, CoreStack</p>
71     </div>
72     <div class="testimonial-card">
73         <p class="quote">"Our site finally feels like it belongs to a real brand. Every
        screen size looks intentional now."</p>
74         <p class="author">– Meera Das, Lumen Retail</p>
75     </div>
76 </div>
77 </section>
78
79 <!-- ===== CONTACT FORM ===== -->
80 <section class="contact" id="contact">
81     <h2>Let's Talk</h2>
82     <form>
83         <label for="name">Name</label>
84         <input type="text" id="name" name="name" placeholder="Your name" required>
85
86         <label for="email">Email</label>
87         <input type="email" id="email" name="email" placeholder="you@example.com" required>
88
89         <label for="message">Message</label>
90         <textarea id="message" name="message" rows="5" placeholder="Tell us about your
        project..." required></textarea>
91
92         <button type="submit" class="btn">Send Message</button>
93     </form>
94 </section>
95
96 <!-- ===== FOOTER ===== -->
97 <footer>
98     <p>&copy; 2026 PixelForge Studio. All rights reserved.</p>
99 </footer>
100
101 <script>
102     // Sticky header shadow on scroll
103     const header = document.getElementById('site-header');
104     window.addEventListener('scroll', () => {
105         header.classList.toggle('scrolled', window.scrollY > 20);
106     });
107
108     // Fade-in on scroll for sections and cards
109     const revealItems = document.querySelectorAll('.card, .testimonial-card, .about,
        .contact');
110     const observer = new IntersectionObserver((entries) => {
111         entries.forEach(entry => {
112             if (entry.isIntersecting) {
113                 entry.target.classList.add('reveal-visible');
114                 observer.unobserve(entry.target);
115             }
116         });
117     }, { threshold: 0.15 });
118     revealItems.forEach(item => {
119         item.classList.add('reveal');

```

120	observer.observe(item);
121	});
122	</script>
123	</body>
124	</html>

• • • week-3.css

```
1  /* ===== RESET & BASE ===== */
2  * {
3      margin: 0;
4      padding: 0;
5      box-sizing: border-box;
6  }
7  html {
8      scroll-behavior: smooth;
9  }
10 body {
11     font-family: 'Segoe UI', Arial, sans-serif;
12     line-height: 1.6;
13     color: #2d2a3d;
14 }
15 a {
16     text-decoration: none;
17     color: inherit;
18 }
19 ul {
20     list-style: none;
21 }
22
23 /* ===== NAVIGATION BAR ===== */
24 header {
25     background-color: #2b1e4a;
26     position: sticky;
27     top: 0;
28     z-index: 100;
29     transition: box-shadow 0.3s ease, background-color 0.3s ease;
30 }
31 header.scrolled {
32     box-shadow: 0 4px 15px rgba(0, 0, 0, 0.3);
33     background-color: #23163d;
34 }
35 nav {
36     display: flex;
37     justify-content: space-between;
38     align-items: center;
39     max-width: 1100px;
40     margin: 0 auto;
41     padding: 15px 20px;
42     flex-wrap: wrap;
43 }
44 .logo {
45     font-size: 1.5rem;
46     font-weight: bold;
47     background: linear-gradient(90deg, #ff7a59, #ffb347);
48     -webkit-background-clip: text;
49     background-clip: text;
50     color: transparent;
51 }
52 .nav-links {
53     display: flex;
54     gap: 25px;
```

```

55 }
56 .nav-links a {
57     color: #e4defc;
58     font-weight: 500;
59     position: relative;
60     padding-bottom: 4px;
61 }
62 .nav-links a::after {
63     content: '';
64     position: absolute;
65     left: 0;
66     bottom: 0;
67     width: 0%;
68     height: 2px;
69     background-color: #ff7a59;
70     transition: width 0.3s ease;
71 }
72 .nav-links a:hover {
73     color: #ff7a59;
74 }
75 .nav-links a:hover::after {
76     width: 100%;
77 }
78
79 /* ===== HERO SECTION ===== */
80 .hero {
81     background: linear-gradient(135deg, #2b1e4a 0%, #4a2f7a 55%, #6a3fa0 100%);
82     color: #fff;
83     text-align: center;
84     padding: 110px 20px;
85 }
86 .hero h1 {
87     font-size: 2.7rem;
88     margin-bottom: 15px;
89     animation: fadeInUp 0.8s ease both;
90 }
91 .hero p {
92     font-size: 1.1rem;
93     max-width: 600px;
94     margin: 0 auto 30px;
95     color: #e4defc;
96     animation: fadeInUp 0.8s ease 0.15s both;
97 }
98 .hero .btn {
99     animation: fadeInUp 0.8s ease 0.3s both;
100 }
101 .btn {
102     display: inline-block;
103     background: linear-gradient(90deg, #ff7a59, #ffb347);
104     color: #2b1e4a;
105     padding: 12px 30px;
106     border-radius: 30px;
107     font-weight: bold;
108     border: none;
109     cursor: pointer;
110     font-size: 1rem;

```

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111     transition: transform 0.25s ease, box-shadow 0.25s ease;
112 }
113 .btn:hover {
114     transform: translateY(-3px);
115     box-shadow: 0 8px 20px rgba(255, 122, 89, 0.4);
116 }
117
118 /* ===== SCROLL REVEAL ANIMATION ===== */
119 .reveal {
120     opacity: 0;
121     transform: translateY(25px);
122     transition: opacity 0.6s ease, transform 0.6s ease;
123 }
124 .reveal-visible {
125     opacity: 1;
126     transform: translateY(0);
127 }
128
129 /* ===== ABOUT SECTION ===== */
130 .about {
131     max-width: 900px;
132     margin: 0 auto;
133     padding: 80px 20px;
134     text-align: center;
135 }
136 .about h2 {
137     font-size: 2rem;
138     margin-bottom: 20px;
139     color: #2b1e4a;
140 }
141 .about p {
142     color: #5b5470;
143 }
144
145 /* ===== SERVICES SECTION ===== */
146 .services {
147     background-color: #f5f1ff;
148     padding: 80px 20px;
149     text-align: center;
150 }
151 .services h2 {
152     font-size: 2rem;
153     margin-bottom: 50px;
154     color: #2b1e4a;
155 }
156 .service-cards {
157     display: flex;
158     flex-wrap: wrap;
159     justify-content: center;
160     gap: 30px;
161     max-width: 1100px;
162     margin: 0 auto;
163 }
164 .card {
165     background-color: #fff;
166     border-radius: 12px;

```



```

167     padding: 35px 25px;
168     flex: 1 1 280px;
169     max-width: 320px;
170     box-shadow: 0 4px 10px rgba(43, 30, 74, 0.08);
171     border-top: 4px solid #ff7a59;
172     transition: transform 0.3s ease, box-shadow 0.3s ease;
173 }
174 .card-icon {
175     font-size: 2rem;
176     margin-bottom: 12px;
177 }
178 .card:hover {
179     transform: translateY(-8px);
180     box-shadow: 0 12px 24px rgba(43, 30, 74, 0.16);
181 }
182 .card h3 {
183     margin-bottom: 12px;
184     color: #2b1e4a;
185 }
186 .card p {
187     color: #6b6480;
188     font-size: 0.95rem;
189 }
190
191 /* ===== TESTIMONIALS SECTION ===== */
192 .testimonials {
193     padding: 80px 20px;
194     text-align: center;
195     background-color: #fff;
196 }
197 .testimonials h2 {
198     font-size: 2rem;
199     margin-bottom: 50px;
200     color: #2b1e4a;
201 }
202 .testimonial-cards {
203     display: flex;
204     flex-wrap: wrap;
205     justify-content: center;
206     gap: 30px;
207     max-width: 1100px;
208     margin: 0 auto;
209 }
210 .testimonial-card {
211     background: linear-gradient(160deg, #2b1e4a, #4a2f7a);
212     color: #e4defc;
213     border-radius: 12px;
214     padding: 30px 25px;
215     flex: 1 1 280px;
216     max-width: 320px;
217     text-align: left;
218     transition: transform 0.3s ease;
219 }
220 .testimonial-card:hover {
221     transform: translateY(-6px);
222 }

```

```

223 .testimonial-card .quote {
224     font-size: 0.95rem;
225     color: #e4defc;
226     margin-bottom: 15px;
227     font-style: italic;
228 }
229 .testimonial-card .author {
230     font-weight: 600;
231     color: #ffb347;
232     font-size: 0.9rem;
233 }
234
235 /* ===== CONTACT FORM ===== */
236 .contact {
237     max-width: 600px;
238     margin: 0 auto;
239     padding: 80px 20px;
240 }
241 .contact h2 {
242     text-align: center;
243     font-size: 2rem;
244     margin-bottom: 30px;
245     color: #2b1e4a;
246 }
247 .contact form {
248     display: flex;
249     flex-direction: column;
250     gap: 15px;
251 }
252 .contact label {
253     font-weight: 600;
254     color: #4a4460;
255 }
256 .contact input,
257 .contact textarea {
258     width: 100%;
259     padding: 12px;
260     border: 1px solid #d8d1ec;
261     border-radius: 8px;
262     font-family: inherit;
263     transition: border-color 0.25s ease, box-shadow 0.25s ease;
264 }
265 .contact input:focus,
266 .contact textarea:focus {
267     outline: none;
268     border-color: #ff7a59;
269     box-shadow: 0 0 3px rgba(255, 122, 89, 0.2);
270 }
271 .contact button {
272     align-self: center;
273 }
274
275 /* ===== FOOTER ===== */
276 footer {
277     background-color: #2b1e4a;
278     color: #cfc6ea;

```

```

279     text-align: center;
280     padding: 25px 20px;
281 }
282
283 /* ===== KEYFRAMES ===== */
284 @keyframes fadeInUp {
285     from {
286         opacity: 0;
287         transform: translateY(20px);
288     }
289     to {
290         opacity: 1;
291         transform: translateY(0);
292     }
293 }
294
295 /* ===== RESPONSIVE MEDIA QUERIES ===== */
296 @media (max-width: 768px) {
297     nav {
298         flex-direction: column;
299         gap: 10px;
300     }
301     .nav-links {
302         flex-wrap: wrap;
303         justify-content: center;
304         gap: 15px;
305     }
306     .hero h1 {
307         font-size: 2.1rem;
308     }
309     .testimonial-cards {
310         gap: 20px;
311     }
312 }
313
314 @media (max-width: 480px) {
315     .hero {
316         padding: 60px 15px;
317     }
318     .about, .services, .contact, .testimonials {
319         padding: 50px 15px;
320     }
321     .card, .testimonial-card {
322         flex: 1 1 100%;
323         max-width: 100%;
324     }
325     .nav-links {
326         gap: 10px;
327         font-size: 0.9rem;
328     }
329 }

```

7. CODE EXPLANATION

Tag / Property	Description
----------------	-------------

<header>, <nav>	Hold the site logo and the navigation links at the top of the page; the header stays sticky while scrolling.
<section id="...">	Marks the hero, about, services, testimonials, and contact blocks so nav links can jump to them.
<div class="service-cards">	A Flexbox container that lays the three service cards out in a row.
<div class="testimonial-cards">	A Flexbox container that lays the three testimonial cards out in a row.
<form>, <label>, <input>, <textarea>	Build the contact form and its labelled fields.
required	HTML5 built-in validation that blocks submission if a field is left empty.
* { box-sizing: border-box; }	Makes padding and borders count inside an element's declared width, avoiding layout surprises.
display: flex; flex-wrap: wrap;	Arranges nav links, service cards, and testimonial cards in a row that wraps onto new lines when needed.
:hover, transform	Applies a lift/colour change only while the mouse pointer is over a link, button, or card.
scroll-behavior: smooth;	Makes clicking a nav link scroll smoothly to its section instead of jumping instantly.
position: sticky; + scroll listener	Keeps the header fixed at the top and adds a shadow once the page is scrolled, via a small JavaScript snippet.
IntersectionObserver	JavaScript API that detects when a section or card enters the viewport and reveals it with a fade/slide animation.
linear-gradient(), box-shadow, border-radius	Give the header, hero, buttons, and cards their gradient colour and soft, rounded, elevated appearance.
@media (max-width: 768px)	Applies alternate CSS rules on tablets and smaller screens.
@media (max-width: 480px)	Applies further adjustments for small mobile screens.

8. TEST CASES AND EXECUTION DATA

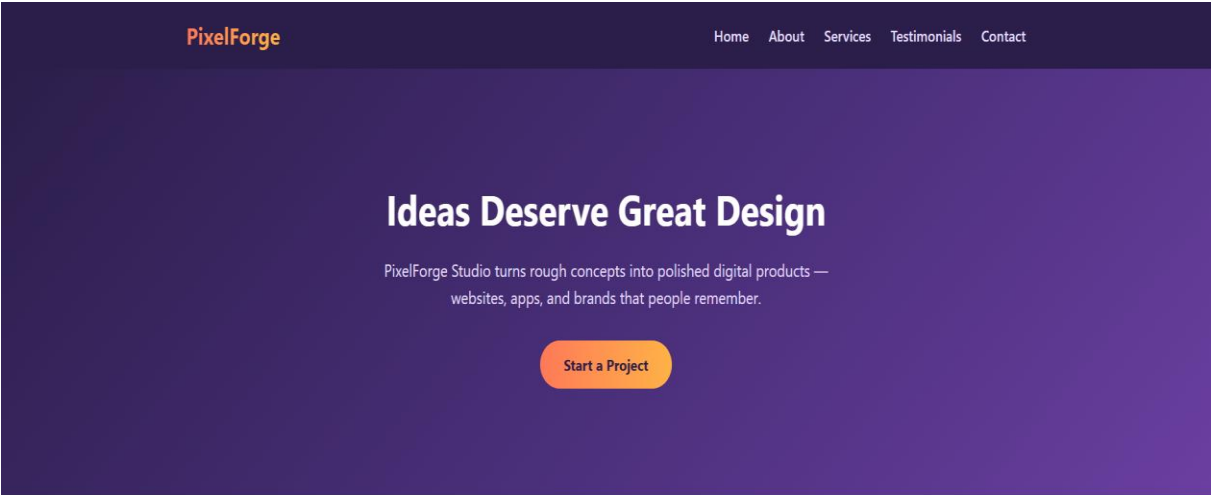
The page was opened in a browser at different window widths, and each interactive element was tested; the results were recorded below.

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Open index.html in a browser	Page loads with header, hero, about, services, testimonials, contact, and footer	Page displayed correctly	Pass
TC-02	Click 'Services' in the navigation bar	Browser scrolls smoothly to the Services section	Smooth scroll worked correctly	Pass
TC-03	Hover over a service card	Card lifts slightly with a stronger shadow	Hover effect displayed	Pass
TC-04	Scroll down towards the Testimonials section	Cards fade and slide into view	Reveal animation worked	Pass
TC-05	Leave the Name field empty and click Send Message	Browser blocks submission and prompts for the field	Required-field prompt shown	Pass
TC-06	Enter an invalid email format	Browser shows an email-format validation error	Validation error displayed	Pass
TC-07	Resize browser window to tablet width ($\leq 768\text{px}$)	Nav links wrap and centre; layout stays readable	Layout adjusted correctly	Pass
TC-08	Resize browser window to mobile width ($\leq 480\text{px}$)	Service and testimonial cards stack in a single column	Cards stacked correctly	Pass

TC-09	Click into the Name input field	Border colour changes to accent colour on focus	Pass	Pass
TC-10	Click 'Start a Project' hero button	Browser scrolls smoothly to the Contact section	Pass	Pass

9. OUTPUT

Output 1: Header, Navigation, and Hero Section (Desktop)



Output 2: About, Services, and Testimonials Sections (Desktop)

Who We Are

PixelForge Studio is a small, focused creative team that partners with startups and growing businesses to design and build products worth using. We obsess over the details — layout, motion, speed, and clarity — so your users never have to think twice.

What We Do



Brand & UI Design

Distinct visual identities and interfaces that reflect who you are, not a generic template.



Web & App Development

Fast, reliable builds using clean code — from landing pages to full product dashboards.



Growth & Optimization

SEO-friendly structure, performance tuning, and conversion-focused layouts that actually convert.

Client Stories

"PixelForge rebuilt our entire product page and bookings jumped almost immediately. They actually listen."

— Priya Nair, Founder, Trailmate

"Clear communication, sharp design instincts, and they delivered ahead of schedule. Rare combination."

— Karthik Iyer, CoreStack

"Our site finally feels like it belongs to a real brand. Every screen size looks intentional now."

— Meera Das, Lumen Retail

Output 3: Contact Form and Footer (Desktop)

Let's Talk

Name

Your name

Email

you@example.com

Message

Tell us about your project...

Send Message

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Output 4: Full Page on a Mobile Screen – Responsive Layout

Ideas Deserve Great Design

PixelForge Studio turns rough concepts into polished digital products — websites, apps, and brands that people remember.

[Start a Project](#)

Who We Are

PixelForge Studio is a small, focused creative team that partners with startups and growing businesses to design and build products worth using. We obsess over the details — layout, motion, speed, and clarity — so your users never have to think twice.

What We Do



Brand & UI Design

Distinct visual identities and interfaces that reflect who you are, not a generic template.



Web & App Development

Fast, reliable, build-once, reuse code.

10. RESULT

Thus, a modern, responsive webpage was successfully designed and developed using HTML5 and CSS3. The sticky navigation bar, hero section, about section, service cards, testimonials section, contact form, and footer were implemented with hover effects, smooth scrolling, and scroll-based reveal animations, and the layout was verified to adapt correctly across desktop, tablet, and mobile screen widths using CSS media queries.